Dhruvil Bhingradiya

Full-Stack & Product Developer • Web, Mobile & Distributed Systems

Remote (Full US ET/PT Overlap)
3+ Years Professional Exp. • Extreme Ownership

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Live Products: tool-web.dev

EXECUTIVE PROFILE

Production Focus • Extreme Ownership

Full-Stack Product Developer with **3+ years of professional engineering experience** taking software systems from blank canvases to production deployment. Proven track record architecting complex multi-role digital products, including a **complete LMS & video-streaming infrastructure (Lux Trader)**, a **3-sided marketplace & dynamic booking platform**, and a **client-side WebAssembly utility suite (ToolWeb)**. Operates with strict Extreme Ownership: rapidly mastering unfamiliar codebases, decomposing ambiguous stakeholder requests into resilient technical specifications, and delivering hardened web and mobile applications with clear technical documentation. Daily practitioner of high-leverage **AI-assisted engineering** (Claude Code, Cursor, Copilot, Grok)—enforcing prompt decomposition, AST verification, and automated regression testing to ship clean, regression-free code at 3× velocity.

CORE TECHNICAL COMPETENCIES

Full-Stack & Systems Architecture

Frontend & Web:	TypeScript, JavaScript (ES6+), React, TanStack Query/Router, HTML5/CSS3, Tailwind CSS, WebAssembly (Rust), Web APIs
Mobile & Cross-Platform:	Flutter & Dart (Android & iOS), Local-First Architecture, SQLite, State Management (Riverpod/Provider), FCM Push Notifications
Backend & Cloud:	Node.js, Express, RESTful APIs, Firebase (Auth, Firestore, Cloud Functions, Storage), Supabase, PostgreSQL, SQLite
Architecture & Media:	Multi-Sided Marketplaces, LMS & Training Systems, FFmpeg Video Transcoding, HLS Multi-Bitrate Streaming, Multi-Role RBAC
Payments & Integrations:	Dynamic Payment Architectures, Razorpay & Stripe Checkout, Webhooks, Third-Party REST Integrations, Inventory/Booking Locks
AI-Assisted Workflow:	Claude Code, Cursor, Antigravity, Grok, GitHub Copilot (Prompt chaining, AST refactoring, automated test synthesis, verification harnesses)

FEATURED PRODUCTION SYSTEMS

Architected & Delivered End-to-End

Lux Trader — Production Trading LMS & Video Infrastructure

Full-Stack Product Developer — Client Delivery • Live Ecosystem

LMS / Training Platform | Flutter Mobile App | Node.js Backend & Admin API | FFmpeg / HLS Video Pipeline | Firebase & Cloud Functions | Razorpay Dynamic Payments

- **End-to-End System Architecture:** Engineered a complete learning management ecosystem comprising a production Flutter mobile application, an administrative web dashboard, and an automated Node.js video processing backend.
- **Adaptive Video Transcoding Pipeline:** Built a dedicated Node.js microservice utilizing FFmpeg to ingest raw master lectures and transcode them into multi-resolution HLS (HTTP Live Streaming) streams, eliminating buffer stalls on low-bandwidth connections.
- **Modular LMS Curriculum & Analytics:** Implemented structured course hierarchies, lesson progress tracking, resume-where-left-off playback state, daily educational tips, and student engagement telemetry for platform administrators.
- **Dynamic Zero-Downtime Payments:** Designed an administrative payment configuration module that dynamically routes and updates Razorpay payment gateways and course pricing from the admin panel without requiring mobile client app redeployments.
- **Operations & Plant Handoff:** Produced comprehensive operational documentation, database schemas, and clean API contracts, enabling non-technical plant and office personnel to manage video assets, courses, and student permissions independently.

Cinema & Service Multi-Role Marketplace Platform

Platform Architect & Lead Developer — Multi-Role Commercial Product

Multi-Sided Marketplace | 3 Distinct Roles (Buyer / Vendor / Super Admin) | Dynamic Seating Engine | Tiered Pricing Matrices | Reservation Lock Engine | Digital QR Ticketing

- **Three-Role Marketplace Ecosystem:** Architected a multi-sided marketplace connecting end customers, service vendors (cinema operators), and platform super-administrators with isolated role-based access control (RBAC).
- **Visual Territory & Hall Builder:** Developed a drag-and-configure interactive visual engine enabling vendor operators to build custom layout grids, row presets, and seating categories tailored to varying venue capacities.
- **Concurrency & Optimistic Locking:** Implemented a collision-free reservation session manager with temporary checkout locks to prevent duplicate bookings during peak-demand flash traffic spikes.
- **Dynamic Surcharge & Pricing Engine:** Formulated flexible pricing algorithms accommodating base category rates, showtime percentage adjustments, weekend surcharges, and platform commission fee deductions.
- **Operator & Platform Dashboards:** Delivered real-time administrative dashboards tracking occupancy rates, gross revenue, category performance, and automated digital QR ticket/pass generation.

FEATURED PRODUCTION SYSTEMS (CONTINUED)

Performance & Client-Side Engineering

ToolWeb — Privacy-First Web Utility & Compute Platform

Creator & Lead Engineer

tool-web.dev • Production Live

WebAssembly (Rust Engine) | TypeScript / ESNext | Client-Side Compute | 20+ Developer Tools | Zero Data Egress

- **Client-Side High-Performance Architecture:** Built a browser-native utility suite of 20+ specialized developer and file tools (image compressor, JSON-to-Dart generator, AST transformers, regex tester) operating entirely within the user's browser sandbox.
- **Rust & WebAssembly Computational Core:** Compiled compute-heavy processing modules (image compression, binary format conversions, complex parsers) to WebAssembly via Rust, slashing execution times to near-native speeds.
- **Developer Format Converters:** Implemented AST-level transformation tools including JSON-to-Dart model generators and HTML-to-Flutter converter pipelines with instantaneous client-side preview rendering.
- **Zero-Cost & Privacy-Compliant:** Eliminated server bandwidth and processing egress costs while guaranteeing 100% data privacy for confidential documents and proprietary schema snippets.

Production Mobile & Local-First Systems: StyleCalc & TaskFlow

Mobile Engineer

Google Play Store • Open Architecture

Google Play Live | Flutter & Dart | Offline-First Local Storage | Hardware Background Alarms | Zero-Crash Metric

- **Published Product Reliability (StyleCalc):** Shipped a custom-styled calculator utility to Google Play, maintaining a 100% crash-free user session metric with instantaneous touch-response rendering and full theme persistence.
- **Local-First Scheduling Engine (TaskFlow):** Designed an offline-first productivity application featuring stateful task life-cycles (scheduled, completed, overdue alert states), persistent local alarm scheduling, and on-device weekly performance telemetry.
- **Defensive Mobile Architecture:** Structured declarative Flutter widget trees with granular state isolation to eliminate redundant view rebuilds, optimizing battery and CPU utilization on resource-constrained devices.

PROFESSIONAL EXPERIENCE & DELIVERY TRACK RECORD

Extreme Ownership & Execution

Full-Stack & Product Developer

Independent Engineering & Client Delivery

2022 – Present (3+ Years) • Remote

- **Autonomous Product Delivery:** Acted as end-to-end technical owner for businesses and startups—scoping technical specifications from unstructured client conversations, establishing architecture diagrams, and delivering production web/mobile apps.
- **Rapid Adoption of Unfamiliar Tech Stacks:** Consistently audited, diagnosed, and extended third-party codebases and APIs, resolving legacy performance bottlenecks and deploying mission-critical integrations under tight sprint deadlines.
- **Clear Communication & Extreme Ownership:** Provided structured async status memos, video walkthroughs, and detailed maintenance documentation to cross-functional stakeholders, ensuring zero ambiguity and seamless operational handoffs.
- **Production Reliability & Quality:** Maintained automated build verification pipelines, strict semantic versioning, and defensible error budgets, ensuring zero regressions on live deployed services.
- **US Eastern / Pacific Timezone Alignment:** Maintained consistent 4+ hour daily overlap with US business hours, fostering frictionless collaborative workflows with North American clients and engineering teams.

ENGINEERING METHODOLOGY: HIGH-LEVERAGE AI-ASSISTED CODING

Tested • Verified • Non-Generic

> **Philosophy:** AI coding tools are not an auto-pilot—they are a high-speed precision instrument. True velocity comes from expert prompt decomposition, strict compiler verification, AST validation, and aggressive automated test synthesis to eliminate hallucinations and regressions.

- **Daily Toolchain:** Expert proficiency with **Claude Code, Cursor, Copilot, Grok, and Antigravity** to accelerate boilerplate implementation, schema drafting, and complex API wrapper construction.
- **Verification & Defect Isolation:** Rigorously inspect and refactor all AI-generated code against strict TypeScript rules, Dart static analysis, and defensive runtime error-handling patterns before merging.
- **Automated Testing & Documentation:** Utilize LLMs to generate exhaustive unit test suites covering boundary conditions and to maintain living documentation schemas for plant, warehouse, and non-technical administrative teams.

EDUCATION, FOUNDATIONS & ENGAGEMENT

Qualifications & Availability

Computer Science:	Strong foundations in Data Structures, Object-Oriented Architecture, Relational/NoSQL Database Design, and Systems Security
Work Ethic & Values:	**Extreme Ownership:** Total responsibility from concept to post-release monitoring; zero finger-pointing, fast turnaround
Availability:	Full-time ongoing remote engagement; overlap with US Eastern (ET) / Pacific (PT) timezones; immediate availability